

BRAC University
Department of Electrical & Electronic Engineering

Course: EEE-103 (Computer Programming)

Section: 02

Semester: Summer 2026

Assignment 1

Instructions:

- Answer all questions carefully.
- For coding questions, write proper, well-indented C programs.
- If any given code contains logical or syntax errors, do **not** fix them unless explicitly asked to. (Error identification and correction may also be part of the task.)

Q1.

Develop a **flowchart** to determine the result of the following series:

$$1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{n^3}$$

Q2.

Write a **C program** to determine whether a triangle is a right-angled triangle or not, based on the lengths of its three sides entered by the user.

Hint: Use the Pythagoras theorem — for a triangle with sides a , b , and c :

If $c^2 = a^2 + b^2$, then the triangle is right-angled.

You may first find the largest side and then check if this condition holds.

Q3.

A certain grade of steel is graded according to the following conditions:

- (i) Hardness must be greater than 30
- (ii) Carbon content must be less than 0.5
- (iii) Tensile strength must be greater than 2600

The grades are as follows:

- Grade A — all three conditions are met

- Grade B — conditions (i) and (ii) are met
- Grade C — conditions (ii) and (iii) are met
- Grade D — conditions (i) and (iii) are met
- Grade E — only one condition is met
- Grade F — none of the conditions are met

Write a **C program** that asks the user to enter the hardness, carbon content, and tensile strength, and outputs the corresponding grade of steel.

Q4.

Write a **C program** to print the multiplication table for a given number as shown below:

Example: Entered number: 5

```
5 x 1 = 5
...
5 x 10 = 50
```

Q5.

Write a **C program** to find factorials of all numbers between n and m . Here, both n and m are user inputs. Assume always $n < m$.

Q6.

Write a **C program** to find the sum of the following series:

$$\frac{x^1}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + \frac{x^n}{n!}$$

Here, both x and n are user inputs.

Q7.

Write a C program to display the following pattern for the given input. (Use proper loops and logic.)

Input: 5

Output:

```
A
BA
CBA
DCBA
EDCBA
```

Q8.

Find the **output** of the following code (Use a proper tracing table):

```
#include <stdio.h>
int main()
{
    int i = 0;
    for (int i = 1; i < 8; i++)
    {
        if(i == 2)
        {
            break;
        }
        for (int j = 1; j < 3; j++)
        {
            if (j == 2)
                continue;
            printf("%d ", i);
        }
        printf("\n");
    }
    return 0;
}
```

Q9.

Find the **output** of the following code (Use a proper tracing table):

```
#include <stdio.h>
int main()
{
    int n = 1729;
    int count = 0, digit;
    while (n > 0)
    {
        digit = n % 10;
        if (digit == 6 && digit == 7)
        {
            count++;
        }
        n = n / 10;
    }
}
```

```

    printf("n=%d, sum=%d, digit=%d", n, count, digit);
    return 0;
}

```

Q10.

Find the **output** of the following code (Use a proper tracing table):

```

#include <stdio.h>
int main()
{
    int a = -9, b = 21;
    if (a > 0 && b < 0)
        a++;
    else if (a < 0 && b < 0)
        a--;
    else if (a < 0 && b > 0)
        b--;
    else
        b--;
    printf("%d\n", a + b);
    return 0;
}

```

Q11.

The code given below is expected to evaluate the following series:

$$1 + 2^2 + 3^3 + 4^4$$

Trace the code to find all the logical errors and fix them to make it functional. (**Note:** Major modifications are not allowed.)

```

#include <stdio.h>
int main()
{
    int n = 3, p, q, i, j;
    q = 0;
    p = 0;
    for (i = 0; i < n; i++);
    {
        for (j = 0; j < i; j++);
        {

```

```
        q = q + i;
    }
    p = p * q;
}
printf("Ans: %d\n", p);
return 0;
}
```